Google Cloud Professional Data Engineer

8-Day Training Plan · 4 hours/day · 32 hours total · 09:00–13:00 IST

Aligned to the official Google Skills **Professional Data Engineer Certification** path (path 16) and the PDE exam guide

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Programme Overview

Item	Details
Duration	8 days × 4 hours = 32 hours (all in class)
Timing	09:00 – 13:00 IST
Structure	Official course modules (in Google's order) + additional exam-guide topics
Hands-on	31 core labs (28 official path + 3 catalog), plus optional official labs for extra practice
Lab platform	Google Skills (lab credits / subscription per participant)

Legend

Marker	Meaning
★ **Official**	Lab from the official Professional Data Engineer Certification path
◆ **Catalog**	Google Skills catalog lab outside the path (used only where the path has no lab)
Additional	Topic from the PDE exam guide that isn't a module in the official path
Optional	Remaining official labs for participants who want extra practice

Daily Schedule (Days 1–7)

Time	Session	Duration
09:00 – 09:10	Recap	10 min
09:10 – 09:55	Modules – Part A	45 min
09:55 – 10:50	Labs 1 & 2	55 min
10:50 – 11:00	Break	10 min
11:00 – 11:40	Modules – Part B	40 min
11:40 – 12:40	Labs 3 & 4	60 min
12:40 – 13:00	Scenario drill + quiz	20 min

Day 8 Schedule

Time	Session	Duration
09:00 – 09:10	Recap	10 min
09:10 – 09:55	Modules: Knowledge Catalog, Gemini in BigQuery	45 min
09:55 – 11:20	Labs 8.1 – 8.3	85 min
11:20 – 11:30	Break	10 min
11:30 – 12:30	Mock exam (30 questions)	60 min
12:30 – 13:00	Review + exam tips	30 min

Day 1: Foundations & Introduction to Data Engineering

Additional (exam guide): **GCP Core Foundations** – Resource hierarchy, billing; IAM, service accounts; Cloud Storage

Course: Introduction to Data Engineering on Google Cloud

- Course Introduction
- Data Engineering Tasks and Components
- Data Replication and Migration

Hands-on Labs (Core)

#	Lab	Source
1.1	A Tour of Google Cloud Hands-on Labs	★ **Official**
1.2	IAM Custom Roles	◆ **Catalog**
1.3	Loading data into BigQuery	★ **Official**
1.4	Datastream: PostgreSQL Replication to BigQuery	★ **Official**

Day 2: Data Pipeline Patterns

Course: Introduction to Data Engineering on Google Cloud (continued)

- The Extract and Load Data Pipeline Pattern
- The Extract, Load, and Transform Data Pipeline Pattern
- The Extract, Transform, and Load Data Pipeline Pattern
- Automation Techniques
- Course Summary

Hands-on Labs (Core)

#	Lab	Source
2.1	Lakehouse: Qwik Start	★ **Official**
2.2	Create and execute a SQL workflow in Dataform	★ **Official**
2.3	Use Serverless for Apache Spark to Load BigQuery	★ **Official**
2.4	Use Cloud Run Functions to Load BigQuery	★ **Official**

Optional: *Creating a Streaming Data Pipeline for a Real-Time Dashboard with Dataflow* ★

Day 3: Data Lakes & Data Warehouses

Additional (exam guide): **Choosing Storage** – Cloud SQL, AlloyDB, Spanner; Firestore, Memorystore, Bigtable

Course: Build Data Lakes and Data Warehouses on Google Cloud

- Course Introduction
- Introduction to Modern Data Engineering on Google Cloud
- Building a Data Lakehouse with Cloud Storage, Open Formats, and BigQuery
- Modernizing Data Warehouses with BigQuery and Lakehouse
- Advanced Lakehouse Patterns and Data Governance
- Labs and Best Practices
- Course Summary

Hands-on Labs (Core)

#	Lab	Source
3.1	Federated query with BigQuery	★ Official
3.2	Querying external data and Iceberg tables	★ Official
3.3	Getting Started with BigQuery ML	★ Official
3.4	Vector search with BigQuery	★ Official

Day 4: Build a Data Warehouse with BigQuery (Skill Badge)

Additional (exam guide): **BigQuery Deep Dive** – Architecture and slots, pricing and editions; Partitioning and clustering, materialised views; Time travel and snapshots; Authorised views, row- and column-level security

Skill Badge: Build a Data Warehouse with BigQuery

Hands-on Labs (Core)

#	Lab	Source
4.1	Creating a Data Warehouse Through Joins and Unions	★ **Official**
4.2	Creating Date-Partitioned Tables in BigQuery	★ **Official**
4.3	Troubleshooting and Solving Data Join Pitfalls	★ **Official**
4.4	Working with JSON, Arrays, and Structs in BigQuery	★ **Official**

Optional: *Build a Data Warehouse with BigQuery: Challenge Lab ★ (earns the skill badge)*

Day 5: Batch Data Pipelines

Course: Build Batch Data Pipelines on Google Cloud

- When to choose batch data pipelines
- Design and build batch data pipelines
- Control data quality in batch data pipelines
- Orchestrate and monitor batch data pipelines

Additional (exam guide): **Orchestration** – Cloud Composer / Airflow; Workflows, Cloud Scheduler

Hands-on Labs (Core)

#	Lab	Source
5.1	Build a Simple Batch Pipeline with Dataflow Job Builder UI	★ **Official**
5.2	Validate Data Quality for a Batch Data Pipeline using Serverless for Apache Spark	★ **Official**
5.3	Building Batch Pipelines in Cloud Data Fusion	★ **Official**
5.4	Cloud Composer: Qwik Start – Console	◆ **Catalog**

Note: “Use Serverless for Apache Spark to Load BigQuery” is also part of this course; it's already done on Day 2.

Day 6: Dataflow Foundations, Security & Pipeline Development

Course: Serverless Data Processing with Dataflow: Foundations

- Introduction
- Beam Portability
- Separating Compute and Storage with Dataflow
- IAM, Quotas, and Permissions
- Security

Additional (exam guide): **Security & Compliance** – Google-managed, customer-managed (CMEK) and customer-supplied (CSEK) keys; Cloud KMS, Secret Manager; VPC Service Controls; Sensitive Data Protection

Course: Serverless Data Processing with Dataflow: Develop Pipelines (part 1)

- Introduction
- Beam Concepts Review
- Windows, Watermarks, and Triggers
- Sources and Sinks
- Schemas
- State and Timers

Hands-on Labs (Core)

#	Lab	Source
6.1	Setup IAM and Networking for your Dataflow Jobs	★ Official
6.2	Getting Started with Cloud KMS	◆ Catalog
6.3	Writing an ETL Pipeline using Apache Beam and Dataflow (Python)	★ Official
6.4	Batch Analytics Pipelines with Dataflow (Python)	★ Official

Optional: *Branching Pipelines (Python) ★; Sensitive Data Protection de-identification ◆*

Day 7: Streaming Pipelines & Dataflow Operations

Course: Build Streaming Data Pipelines on Google Cloud

- Course Introduction
- Streaming use cases and reference architectures
- Product deep dives
- Key takeaways

Course: Serverless Data Processing with Dataflow: Develop Pipelines (part 2)

- Best Practices
- Dataflow SQL and DataFrames
- Beam Notebooks
- Summary

Course: Serverless Data Processing with Dataflow: Operations

- Monitoring
- Logging and Error Reporting
- Troubleshooting and Debug
- Performance
- Testing and CI/CD
- Reliability
- Flex Templates

Additional (exam guide): **Reliability & Cost** – Disaster recovery (recovery point and time objectives); Quotas, cost controls

Hands-on Labs (Core)

#	Lab	Source
7.1	Stream E-sports Data with Pub/Sub and BigQuery	★ **Official**
7.2	Using Dataflow for Streaming Analytics (Python)	★ **Official**
7.3	Use Apache Beam and Bigtable to Enrich esports DLC Data	★ **Official**
7.4	Monitoring, Logging and Error Reporting for Dataflow Jobs	★ **Official**

Optional: *Stream data with pipelines – Esports Use case ★; Monitor E-sports Chat with Streamlit ★; Advanced Streaming Analytics Pipeline (Python) ★; Testing with Apache Beam (Python) ★; Custom Dataflow Flex Templates (Python) ★*

Day 8: Data Mesh, Gemini in BigQuery & Exam Readiness

Skill Badge: Build a Data Mesh with Knowledge Catalog

- Lakes and zones, aspects, security, data quality

Course: Boost Productivity with Gemini in BigQuery

- Data exploration, code development, data canvas

Course: Work with Gemini Models in BigQuery

- Generative AI with SQL and Python notebooks

Exam Readiness

- Mock exam (30 scenario-based questions)
- Review and exam tips
- Build a Certification Study Guide: PDE Exam Prep (official path) for self-preparation afterwards

Hands-on Labs (Core)

#	Lab	Source
8.1	Knowledge Catalog: Qwik Start – Console	★ **Official**
8.2	Assessing Data Quality with Knowledge Catalog	★ **Official**
8.3	Analyze Customer Reviews with Gemini Using SQL	★ **Official**

Optional: *Create and Add Aspects to Knowledge Catalog Assets ★; Implementing Security in Knowledge Catalog ★; Knowledge Catalog Challenge Lab ★; Explore Data with Gemini in BigQuery ★; Develop Code with Gemini in BigQuery ★; Use Data Canvas to Visualize and Design Queries ★; Analyze Customer Reviews with Gemini Using Python Notebooks ★*

Coverage & Delivery Notes

Official Path Coverage

Official path activity	Day(s)
A Tour of Google Cloud Hands-on Labs	1
Introduction to Data Engineering on Google Cloud	1 – 2
Build Data Lakes and Data Warehouses on Google Cloud	3
Build a Data Warehouse with BigQuery (skill badge)	4
Build Batch Data Pipelines on Google Cloud	5
Serverless Data Processing with Dataflow: Foundations	6
Serverless Data Processing with Dataflow: Develop Pipelines	6 – 7
Build Streaming Data Pipelines on Google Cloud	7
Serverless Data Processing with Dataflow: Operations	7
Build a Data Mesh with Knowledge Catalog (skill badge)	8
Boost Productivity with Gemini in BigQuery	8
Work with Gemini Models in BigQuery	8
Build a Certification Study Guide: PDE Exam Prep	8 (self-prep)

Exam Guide Coverage

Exam section	Day(s)
1. Designing data processing systems	1, 2, 6
2. Ingesting and processing data	2, 5, 6, 7
3. Storing the data	3, 4
4. Preparing and using data for analysis	3, 4, 8
5. Maintaining and automating data workloads	5, 7

Delivery Notes

- **Lab timing:** official labs usually take 45–90 minutes. With about 2 hours of lab time a day, plan for 2 labs done fully and 2 done as guided key tasks.
- **Lab access:** make sure every participant has Google Skills access or credits before Day 1.
- **Naming:** Dataplex appears as “Knowledge Catalog” in the current official path.
- **Catalog labs (◆):** exact titles to be confirmed against the Google Skills catalog.
- **Scenario drills:** a 20-minute exam-style quiz every day (Days 1–7), mixing in questions from earlier days.